

Subhajit Maity

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🐙 [GitHub](#) | 🎓 [Google Scholar](#) | 🆔 [ORCID](#) | 🔗 [LinkedIn](#)

Education

University of Central Florida

Orlando, FL, USA

PhD in Computer Science | **GPA:** 4.0/4.0

Aug. 2024 - Present

Advisor: [Dr. Aritra Dutta](#) | Research Area: Self-Attention, Transformers

Jalpaiguri Government Engineering College (Autonomous)

Jalpaiguri, India

BTech. in Electronics & Communication Engineering | **GPA:** 8.47/10

Aug. 2014 - Jul. 2018

Courses: Linear Algebra, Calculus, Artificial Intelligence

Experiences

University of Central Florida

Orlando, FL, USA

Graduate Research Associate | Skills: Python, PyTorch, Deep Learning, Computer Vision

Aug. 2024 - Present

Supervisor: [Dr. Aritra Dutta](#) | Projects: [Fibottention](#), [KArAt](#), MammoChat, Denoising Attention

- Worked on Visual Large Language Models, **Kolmogorov-Arnold Networks**, Sparse **Self-Attention**.
- Designed the first ever **learnable attentions** for visual understanding tasks that can model complex token relationships and their interactions for better interpretability and explainability.
- Developed the theoretical basis for the **denoising attention** for improved object-level understanding in vision transformers.

Indian Statistical Institute, Kolkata — Technology Innovation Hub (IDEAS)

Kolkata, India

Associate Research Engineer | Skills: Python, PyTorch, Deep Learning, Numpy, Matplotlib

Jun. 2022 - Jul. 2023

Supervisor: [Prof. Umapada Pal](#) | Projects: Traffic Surveillance for Overspeeding Vehicles and License Plate Recognition

- Worked on **Object Detection**, Monocular Speed Estimation, License Plate Recognition, and **Scene Text Recognition**.
- Built a prototype with 30+ fps on NVIDIA RTX 3070 for vehicle speed tracking and identifying license plate registration for overspeed.

Tata Consultancy Services Limited

Kolkata, India

Systems Engineer | Skills: Java, Linux, Oracle OCI, IAM, OAM, OID, FMW SOA Suite, FMW Service Bus

Nov. 2018 - Jan. 2021

Roles: FMW Administrator & Developer, SSO Administrator & Architect, Linux Administrator, OCI Architect

- Designed a cloud server network architecture that **doubled the user load handling capacity** to 5,000+ requests per day.
- Implemented a streamlined role-based access and identity management system for a client with \$15 billion annual revenue. Managed a **team of five people** for administration, development, support, migration, and upgrades in the IAM systems.
- Responded to **four** major incidents for two clients and oversaw recovery with **zero data and revenue loss**.

National Institute of Technology, Silchar

Silchar, India

Research Intern | Skills: MATLAB, Computer Vision, Image Processing

Jun. 2017 - Jul. 2017

Supervisor: [Dr. Koushik Guha](#) | Project: Word Sense Disambiguation for Ambiguous Words using Visual Supervision

- Worked on Image Processing, Natural Language Processing, Bayesian Classifier, and Image Caption Generation.
- Built a classical system for joint language-image correspondences.

Selected Publications & Preprints

Accurate Split Learning on Noisy Signals [[Paper](#)] [[Webpage](#)]

H. Xu, **S. Maity**, A. Dutta, X. Li, P. Kalnis

TMLR

2026

Doodle Your Keypoints: Sketch-Based Few-Shot Keypoint Detection [[Paper](#)] [[Webpage](#)]

S. Maity, A. K. Bhunia, S. Koley, P. N. Chowdhury, A. Sain, Y. Z. Song

ICCV

2025

Sketch Down the FLOPs: Towards Efficient Networks for Human Sketch [[Paper](#)] [[Webpage](#)]

A. Sain, **S. Maity**, P. N. Chowdhury, S. Koley, A. K. Bhunia, Y. Z. Song

CVPR

2025

Kolmogorov-Arnold Attention: Is Learnable Attention Better For Vision Transformers? [[Paper](#)] [[Webpage](#)]

S. Maity, K. Hitsman, X. Li, A. Dutta

Preprint

2025

Fibottention: Inceptive Visual Representation Learning with Diverse Attention Across Heads [[Paper](#)]

A. K. Rahimian, M. K. Govind, **S. Maity**, D. Reilly, C. Kümmerle, S. Das, A. Dutta

Preprint

2024

DistilDoc: Knowledge Distillation for Visually-Rich Document Applications [[Paper](#)]

J. Van Landeghem, **S. Maity**, A. Banerjee, M. Blaschko, M. F. Moens, J. Lladós, S. Biswas

ICDAR

2024

SelfDocSeg: A Self-Supervised vision-based Approach towards Document Segmentation [[Paper](#)] [[Webpage](#)]

S. Maity, S. Biswas, S. Manna, A. Banerjee, J. Lladós, S. Bhattacharya, U. Pal

ICDAR (Oral)

2023

Research Projects

MammoChat: An Anxiety Management Chatbot for Breast Cancer Patients

Orlando, FL, USA

Supervisors: [Dr. Aritra Dutta](#), [Dr. Amrit Singh Bedi](#), [Dr. Yu Tian](#), [Dr. Jane Gibson](#), University of Central Florida

Sep. 2025 - Present

Research Topics: LLMs, Multimodal LLMs, Artificial Emotional Intelligence, Anxiety Management, Emotion Quantification

Denoising Attention for Visual Understanding Tasks (Submitted)		Orlando, FL, USA
Supervisors: Dr. Aritra Dutta , University of Central Florida, Dr. Srijan Das , University of North Carolina Charlotte		Mar. 2025 - Present
Research Topics: Attention, Vision Transformers, Attention Sink, Dispersed Attention, Attention Entropy		
Accurate Split Learning on Noisy Signals (TMLR 2026)		Thuwal, Saudi Arabia
Supervisors: Dr. Aritra Dutta , Dr. Xin Li , University of Central Florida, Dr. Panos Kalnis , King Abdulla University of Science & Technology		May. 2025 - Present
Research Topics: Federated Learning, Split Learning, Signal Denoising in Deep Neural Networks, Differential Privacy		
Learnable Self-Attention for Vision Transformers (Submitted, Preprint Available)		Orlando, FL, USA
Supervisors: Dr. Aritra Dutta , Dr. Xin Li , University of Central Florida		Sep. 2024 - Present
Research Topics: Self-Attention, Vision Transformers, Kolmogorov-Arnold Networks, Kolmogorov-Arnold Representation Theorem		
Fibottention (Preprint Available)		Charlotte, NC, USA
Supervisors: Dr. Aritra Dutta , University of Central Florida, Dr. Christian Kümmerle , Dr. Srijan Das , University of North Carolina Charlotte		Jan. 2024 - Present
Research Topics: Attention, Vision Transformers, Attention Sink, Dispersed Attention, Attention Entropy		
Clouded Leopard Re-Identification, Tracking & Census		Kolkata, India
Collaborators: Dr. Tanoy Mukherjee , Prof. Joydev Chattopadhyay , Indian Statistical Institute, Kolkata, and Mr. Debal Ray , PCCF, Directorate of Forests, The Government of West Bengal, India		Aug. 2023 - Aug. 2024
Research Topics: Feature Detection, Keypoint Detection, Deep Feature Understanding, Point to Point Correspondence		
Sketch-Based Few-Shot Keypoint Detection (ICCV 2025)		Guildford, UK
Collaborators: Dr. Ayan K. Bhunia , Sony Playstation, London, and Prof. Yi-Zhe Song , University of Surrey		Jun. 2023 - Aug. 2024
Research Topics: Meta-Learning, Few-Shot setting, Prototypical Networks, Keypoint Detection		
Knowledge Distillation for Document Applications (ICDAR 2024)		Leuven, Belgium
Collaborators: Dr. Jordy Van Landeghem , Prof. Matthew Blaschko , KU Leuven		May. 2023 - Feb. 2024
Research Topics: Knowledge Distillation, Document Image Classification, Document Layout Analysis		
Self-Supervised Document Layout Analysis (ICDAR 2023)		Barcelona, Spain
Collaborators: Prof. Josep Lladós , Computer Vision Center, Universitat Autònoma de Barcelona		Jan. 2023 - Feb. 2023
Research Topics: Self-supervised Learning, Object Localization, Document Layout Analysis		
Self-Supervised Object Detection		Kolkata, India
Mentors: Prof. Umapada Pal , Indian Statistical Institute, Kolkata, and Prof. Saumik Bhattacharya , Indian Institute of Technology, Kharagpur		Aug. 2022 - Mar. 2023
Research Topics: Self-supervised Learning, Object Localization, Object Detection, Contrastive Learning		
Towards Faster Fine-Grained Sketch-based Image Retrieval (CVPR 2025)		Guildford, UK
Collaborators: Dr. Aneeshan Sain , Sony Playstation, London, and Prof. Yi-Zhe Song , University of Surrey]		Sep. 2021 - Dec. 2022
Research Topics: Knowledge Distillation, Gradient Consensus, Reinforcement Learning, Policy Gradient		
Technical Skills		
Programming	Python, C, MATLAB, Java	
Frameworks	PyTorch, Numpy, Matplotlib, Seaborn, Pandas, Keras, Tensorflow, OpenCV	
Typesetting	L ^A T _E X, Beamer, Microsoft Office	
Subject Expertise	Knowledge Distillation, Large Language Model (LLM), Multi-modal Large Language Model (MLLM), Attention, Dataset Distillation, Self-Supervised Learning, Contrastive Learning, Representation Learning, Generative AI, Agentic AI, Reinforcement Learning with Human Feedback (RLHF), PPO, DPO, Federated Learning, Split Learning	
Honorary & Volunteer Services		
Reviewer		Orlando, FL, USA
Reviewer for CVPR 2026		Nov. 2025
Directorate of Forests, The Government of West Bengal, India		Kolkata, India
Voluntary Technology Consultation for the Leopard Census Project		Sep. 2023
G20 Expo, The Second Education Working Group Meeting, The G20 Summit 2023		Amritsar, India
Representative for India in the Advancement of Cutting-edge Technologies		Mar. 2023